

Dhruv Nigam

MACHINE LEARNING ENGINEER

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Work Experience

Dream 11

Mumbai

LEAD MACHINE LEARNING ENGINEER

July 2022 - Present

- Trained **200M-parameter cross-attention transformer (Lumos)** on **1.7T user tokens** from **250M users**. Replaced **50+ task-specific models**. **+3.15% DAU**, **-2.47% offer spend** in A/B test (700K users/cohort). Built **Accio**, a Delta-native streaming data loader for distributed training on **8x A100s**. [\[tech report\]](#), [\[blog\]](#)
- Built **real-time AI companion** for live cricket streams — **speech-to-speech over WebRTC**, **diffusion-based avatar**, **sub-2s latency**. **1.5M min watchtime**, **1.1M concurrent viewers** at peak (IPL 2026). [\[blog\]](#)
- Built **personalized promotion platform** for **200M users**: supervised model + policy layer + PM tooling. Allocated **INR 2 crore** daily, **zero-downtime mission-critical system**. **+14% WAU**, **100% ROI**.
- Developed **ForeCal**, a probability calibration technique, **published at KDD 2024** [\[paper\]](#). Reduced ECE by **50%**, **1.3% revenue upside**.

Protium

Mumbai

DIRECTOR - DATA SCIENCE AND ENGINEERING

Feb. 2020 - June 2022

- 12th employee and **founding ML Engineer**. Set up the data warehouse by defining the schema and setting up data pipelines built using pyspark to ingest, transform, and load transactional data into Athena. Owned, maintained, and scaled these jobs to **process 200,000+ records daily**.
- Built Deep learning based models using user time series data with 5mn+ records, which beat the gold standard Credit bureau score by 12% in AUC. Build an **industry-first income estimation algorithm using Bayesian modeling**. Together, these models were used to underwrite over **500mn usd worth of loans across India** without human intervention.
- Built the **ML platform** that hosted 52+ models as FastAPI endpoints for **real-time feature engineering and inference** supporting up to 10,000 rpm.

CLSA

Mumbai

QUANTITATIVE TRADER, INVESTMENT BANKING

Nov. 2017 - Feb. 2020

- Founding member hire of a 3-member machine learning team** responsible for **building model-based trade execution strategies** executing over **\$1 bn** in client trades every day at sub 10ms latency. Developed a **Java-based event-driven system** to incorporate real-time machine learning signals into trading decisions.
- Developed and deployed a profitable quantitative trading strategy (Sharpe 1.3) focusing on **cointegration-based time series statistical arbitrage** in the US futures market. The strategy traded over 2 years with over **10 million use in capital** and **4 mn in net profit**.

Open Source, Talks & Writing

Talks ForeCal at **KDD 2024** (Barcelona), LUMOS at **Swiggy** (invited), GPU Fundamentals at **MumbaiPy**.

Open Source Merged PRs in [LiveKit Agents](#) and [TRL](#) (HuggingFace).

Website dhruvnigam93.github.io — talks, blog, and paper notes.

Education

India Institute of Technology - Bombay (IITB)

Mumbai

BACHELORS AND MASTERS IN ELECTRICAL ENGINEERING, SPECIALIZING IN SIGNAL PROCESSING

2011 - 2016

- Master's thesis: [Single Channel Blind Dereverberation of Speech Signals](#)